

6.5830

Lecture 7

Join Algorithms



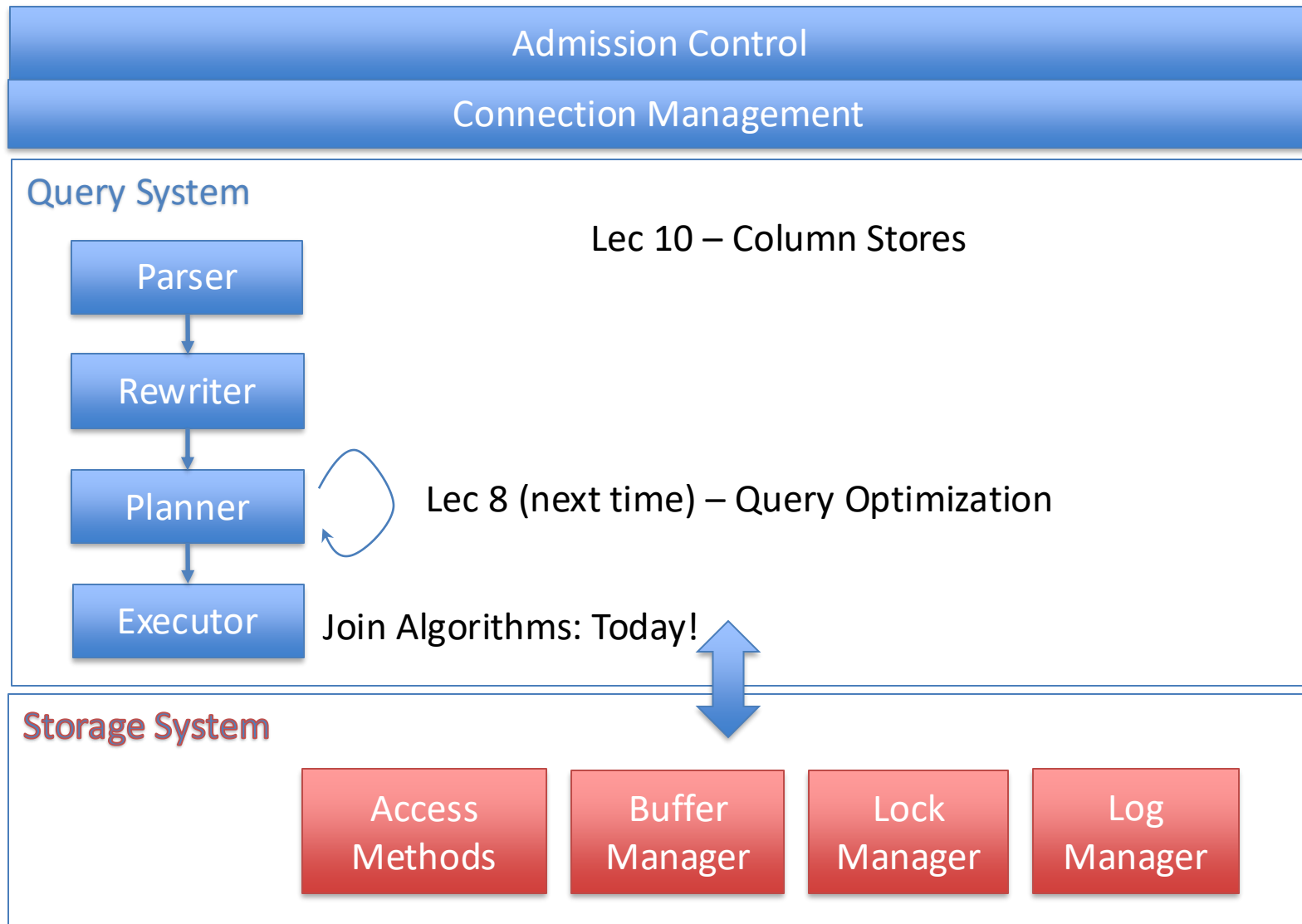
Administration

- Due today: **lab1, group assignment**
- PSet2, Lab 2 already out

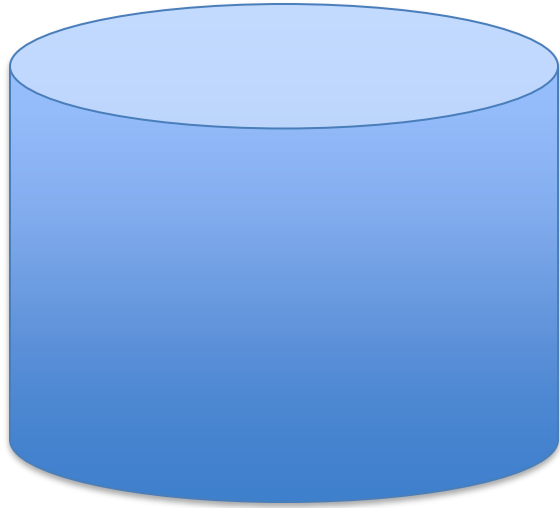
Important dates:

- Lab writeup on March 5 in class; no electronics!
- PSet2, Project Proposals due March 10
- Quiz 1 in-class on March 12
- Lab 2 due March 19
- We will contact you about special arrangements for in-class activities

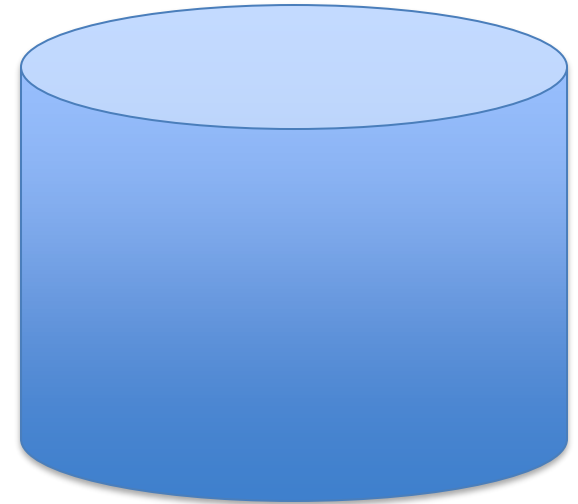
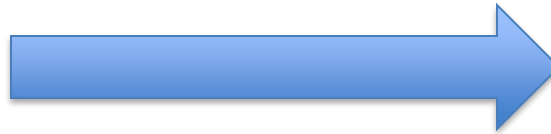
Plan for Next Few Lectures



Typical Database Setup



“Extract, Transform, Load”



Transactional database

Lots of writes/updates

Reads of individual records

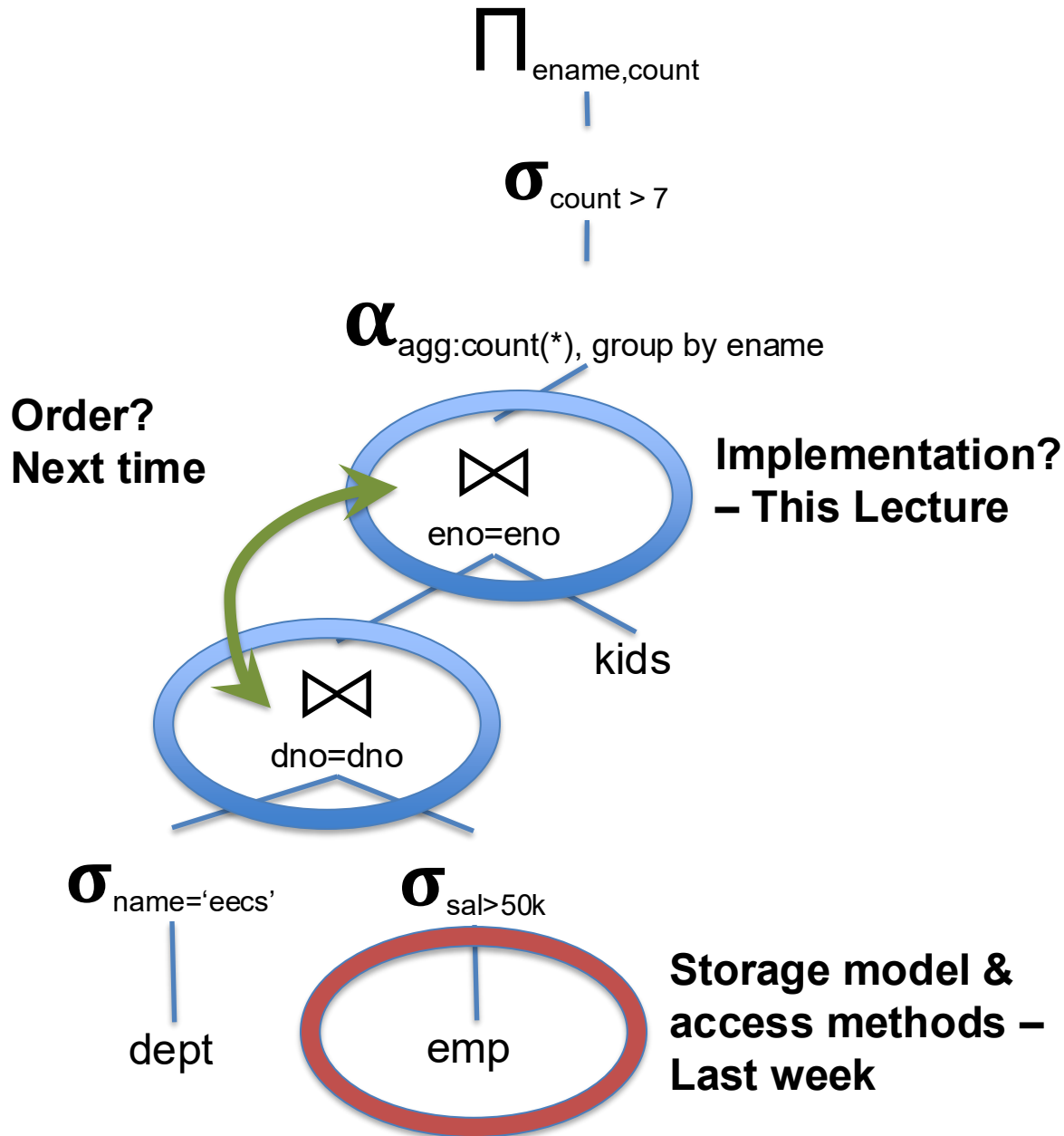
Analytics / Reporting Database “Warehouse”

Lots of reads of many records

Bulk updates

Typical query touches a few columns

Plan questions



Join Algorithms

- Nested loops (NL)
- Blocked nested loops
- Index nested loops (INL)
- When tables fit in memory
 - Hash (only 1 needs to fit)
 - Sort merge (both must fit)
- When tables don't fit into memory
 - Blocked hash join
 - External sort merge
 - Simple hash
 - Grace hash

Notation

Evaluating $\text{Join}(S, R, \text{predicate})$

Assume R is always the smaller table

$\{S\}$ – number of records in S

$|S|$ – number of pages of S

Memory of size M pages

Nested Loops

```
for s in S:  
    for r in R  
        if pred(s,r):  
            output s join r
```

Inner vs outer matters, if only one relation fits in memory

$\{S\} * \{R\}$ comparisons in either case

$|S| + |R|$ IO complexity if everything fits in memory

<https://clicker.mit.edu/6.5830/>

	CPU Complexity	I/O Complexity
(A)	$\{R\} \times S \log \{S\}$	$ S + R $
(B)	$\{R\} \times \{S\}$	$ S + S \times R $
(C)	$\{R\} \times \{S\}$	$ S + \{S\} \times R $
(D)	$ S \times R $	$ S + R $
(E)	$\{R\} \times \{S\}$	$ S + R $
(F)	$\{R\} \times S $	$ S + S \times R $

Select all possible correct solutions

Basic Join Summary

	CPU Complexity	I/O Complexity	Notes
Nested loops	$\{R\} \times \{S\}$	$ S + \{S\} R $ <i>R doesn't fit in memory</i> $ S + R $ <i>R fits in memory</i>	Choice of inner / outer matters when R fits in memory and S doesn't

Block Nested Loops

B = block size ($< M$)

while (not at end of R):

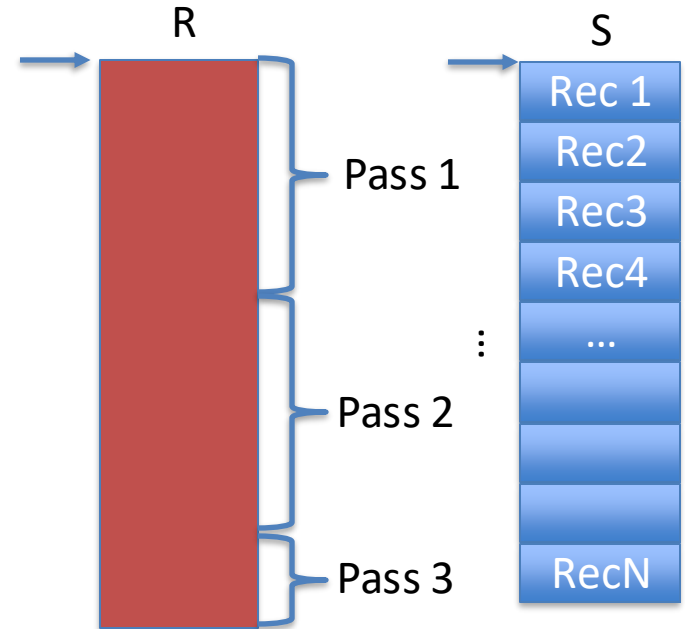
$R' = \text{read } B \text{ records from } R$

 for s in S :

 for r in R' :

 if $\text{pred}(s, r)$:

 output $s \text{ join } r$



Inner vs outer matters; $\{S\} * \{R\}$
comparisons, but $\{R\}/B$ passes over S

<https://clicker.mit.edu/6.5830/>

	CPU Complexity	I/O Complexity
(A)	$\{R/M\} \times \{S\}$	$\lceil R /M \rceil \times S + R/M $
(B)	$\{R\} \times \{S/M\}$	$\lceil R /M \rceil \times S + R $
(C)	$\{R\} \times \{S\}$	$\lceil R /M \rceil \times S + R $
(D)	$\{R\} \times \{S\}$	$\lceil R /M \rceil \times S $

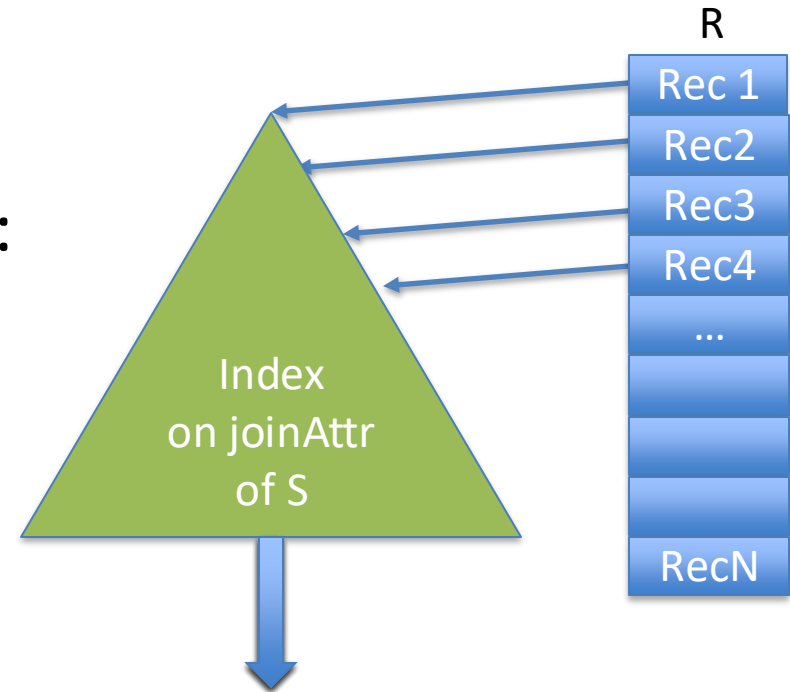
Basic Join Summary

	CPU Complexity	I/O Complexity	Notes
Nested loops	$\{R\} \times \{S\}$	$ S + \{S\} R $ <i>R doesn't fit in memory</i> $ S + R $ <i>R fits in memory</i>	Choice of inner / outer matters when R fits in memory and S doesn't
Blocked nested loops	$\{R\} \times \{S\}$	$\lceil R /M \rceil \times S + R $ <i>Here we use M not B</i>	Better to partition R (fewer passes)

Index Nested Loops

- Assume Index I on Join Attribute of S

```
for r in R:  
    for s in (lookup r.joinAttr in I):  
        output r join s
```



Inner vs outer matters; {R} lookups
Inner is always indexed attribute

Note that index lookups are random, unless R is ordered on join attribute and index is clustered on join attribute

Basic Join Summary

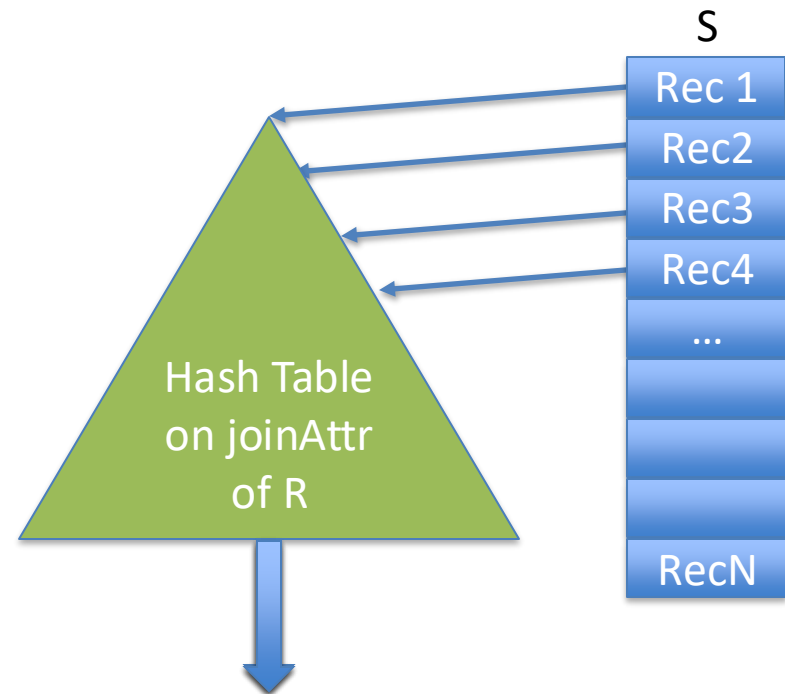
	CPU Complexity	I/O Complexity	Notes
Nested loops	$\{R\} \times \{S\}$	$ S + \{S\} R $ <i>R doesn't fit in memory</i> $ S + R $ <i>R fits in memory</i>	Choice of inner / outer matters when R fits in memory and S doesn't
Blocked nested loops	$\{R\} \times \{S\}$	$\lceil R /M \rceil \times S + R $	Better to partition R (fewer passes)
Index nested loops	$\{R\} \times D$ <i>D is tree depth, $< \sim 5$</i>	$\{R\} \times D$ <i>I/O random unless R sorted & index clustered on join attr</i>	Assuming index on S.

(In Memory) Hash Join

- Essentially the same as index nested loops, with in-memory hash “index” built on the fly
- Build hash table T on join attribute of R

```
T = build hash table on joinAttr of R
for s in S:
    for r in lookup s.joinAttr in T:
        output s join r
```

Inner vs outer matters; {S}
lookups, requires memory to hold
hash table on R



<https://clicker.mit.edu/6.5830/>

	CPU Complexity	I/O Complexity
(A)	$\{R\} * \{S\}$	$ R + 2 * S $
(B)	$\{R\} + \{S\}$	$ R * S $
(C)	$\{R\} * \{S\}$	$ R + S $
(D)	$\{R\} + \{S\}$	$ R + S $

R is the inner table and fits into main memory

Basic Join Summary

	CPU Complexity	I/O Complexity	Notes
Nested loops	$\{R\} \times \{S\}$	$ S + \{S\} R $ <i>R doesn't fit in memory</i> $ S + R $ <i>R fits in memory</i>	Choice of inner / outer matters when R fits in memory and S doesn't
Blocked nested loops	$\{R\} \times \{S\}$	$\lceil R /M \rceil \times S + R $	Better to partition R (fewer passes)
Index nested loops	$\{R\} \times D$ <i>D is tree depth, $< \sim 5$</i>	$\{R\} \times D$ <i>I/O random unless R sorted & index clustered on join attr</i>	Assuming index on S.
Hash join	$\{R\} + \{S\}$	$ R + S $	R must fit into memory

Blocked Hash

- Similar to block nested loops
- Iteratively:
 - Build hash table on chunk of R so that hash table fits in memory
 - Probe (lookup in) with all of S
 - Repeat with next chunk of R

Basic Join Summary

	CPU Complexity	I/O Complexity	Notes
Nested loops	$\{R\} \times \{S\}$	$ S + \{S\} R $ <i>R doesn't fit in memory</i> $ S + R $ <i>R fits in memory</i>	Choice of inner / outer matters when R fits in memory and S doesn't
Blocked nested loops	$\{R\} \times \{S\}$	$\lceil R /M \rceil \times S + R $	Better to partition R (fewer passes)
Index nested loops	$\{R\} \times D$ <i>D is tree depth, $< \sim 5$</i>	$\{R\} \times D$ <i>I/O random unless R sorted & index clustered on join attr</i>	Assuming index on S.
Hash join	$\{R\} + \{S\}$	$ R + S $	Both tables must fit in memory
Blocked hash join	$\{R\} + (\lceil R /M \rceil \times \{S\})$	$\lceil R /M \rceil \times S + R $	

Sort Merge Join

- Sort both S and R (or use index on each to traverse in order)
- Merge (no shared duplicates)

while ($i < \{R\}$ and $j < \{S\}$):

 if ($R[i].\text{joinAttr} == S[j].\text{joinAttr}$):

 output $R[i] \text{ join } S[j]$

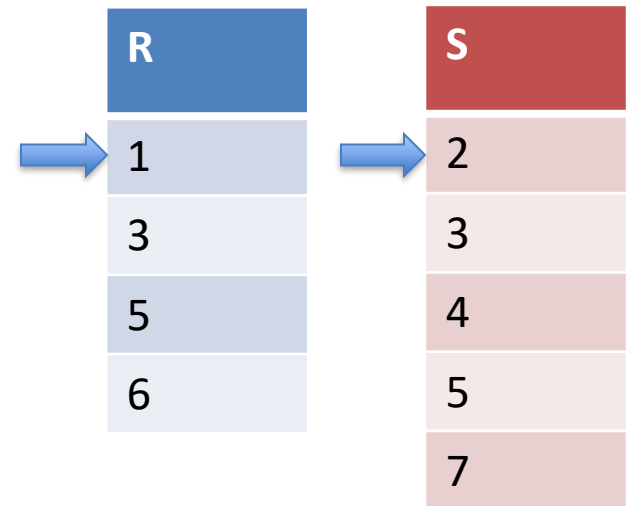
 if ($R[i].\text{joinAttr} < S[j].\text{joinAttr}$):



$i = i + 1$

 else:

$j = j + 1$



Output:

Sort Merge Join

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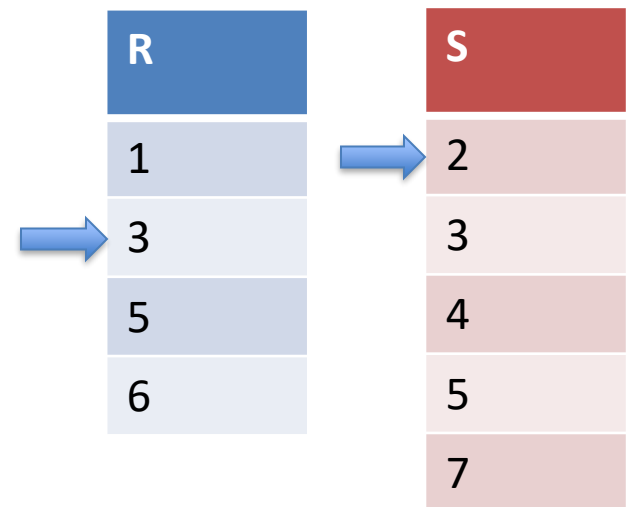
if ($R[i].\text{joinAttr} < S[j].\text{joinAttr}$):

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else:



$j = j + 1$



Output:

Sort Merge Join

- Sort both S and R (or use index on each to traverse in order)
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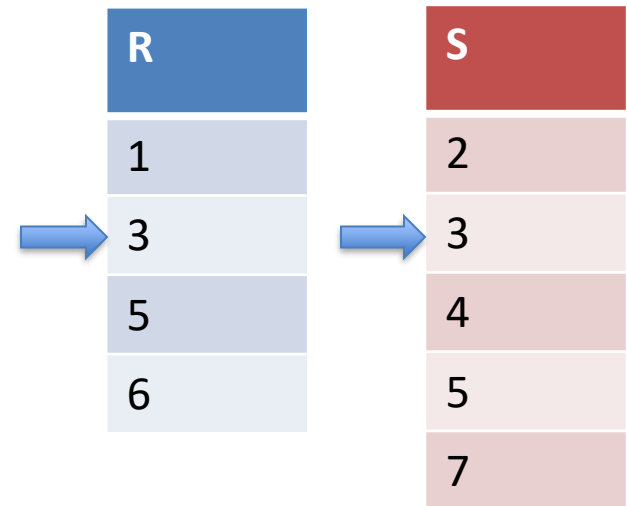
while ($i < \{R\}$ and $j < \{S\}$):

✓ if ($R[i].\text{joinAttr} == S[j].\text{joinAttr}$):
 output $R[i] \text{ join } S[j]$

 if ($R[i].\text{joinAttr} < S[j].\text{joinAttr}$):
 $i = i + 1$

 else:

✓ $j = j + 1$



Output: 3

Sort Merge Join

- Sort both S and R (or use index on each to traverse in order)
- Merge (no shared duplicates)

while ($i < \{R\}$ and $j < \{S\}$):

if ($R[i].\text{joinAttr} == S[j].\text{joinAttr}$):

output $R[i]$ join $S[j]$

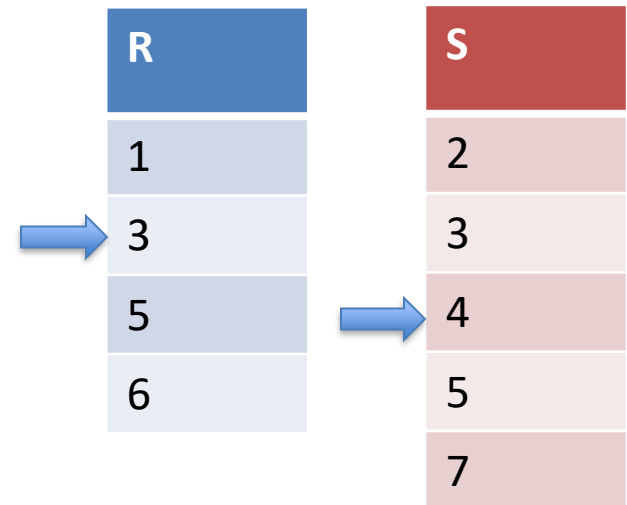
if ($R[i].\text{joinAttr} < S[j].\text{joinAttr}$):



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Output: 3

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while ($i < \{R\}$ and $j < \{S\}$):

if ($R[i].\text{joinAttr} == S[j].\text{joinAttr}$):

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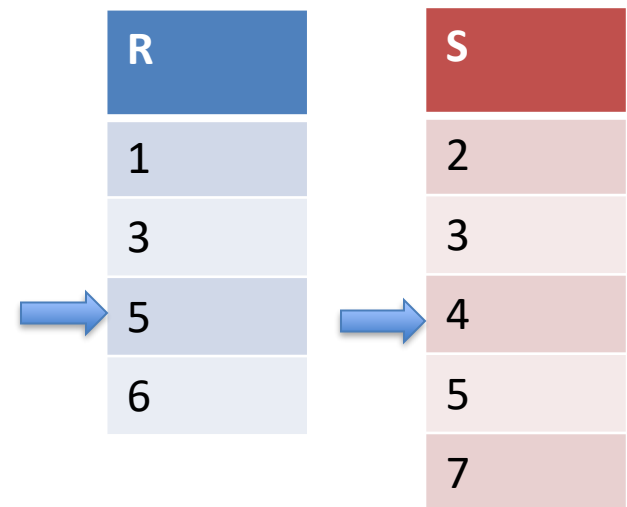
if ($R[i].\text{joinAttr} < S[j].\text{joinAttr}$):

$i = i + 1$

else:



$j = j + 1$



Output: 3

Sort Merge Join

- Sort both S and R (or use index on each to traverse in order)
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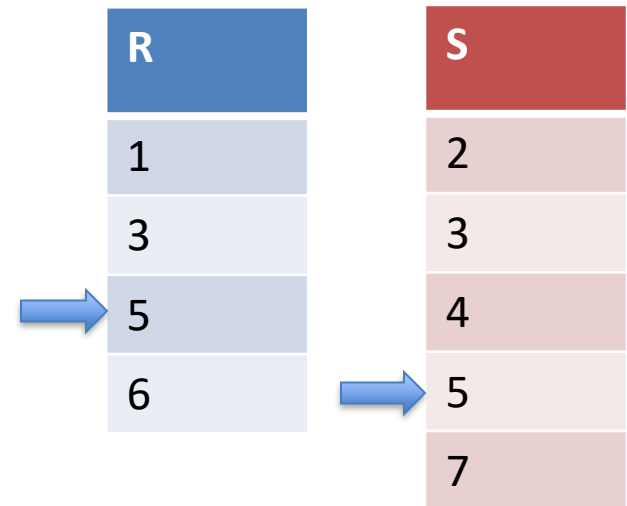
while ($i < \{R\}$ and $j < \{S\}$):

✓ if ($R[i].\text{joinAttr} == S[j].\text{joinAttr}$):
 output $R[i]$ join $S[j]$

 if ($R[i].\text{joinAttr} < S[j].\text{joinAttr}$):
 $i = i + 1$

 else:

✓ $j = j + 1$



Output: 3, 5

Sort Merge Join

- Sort both S and R (or use index on each to traverse in order)
- Merge (no shared duplicates)

while ($i < \{R\}$ and $j < \{S\}$):

if ($R[i].\text{joinAttr} == S[j].\text{joinAttr}$):

output $R[i]$ join $S[j]$

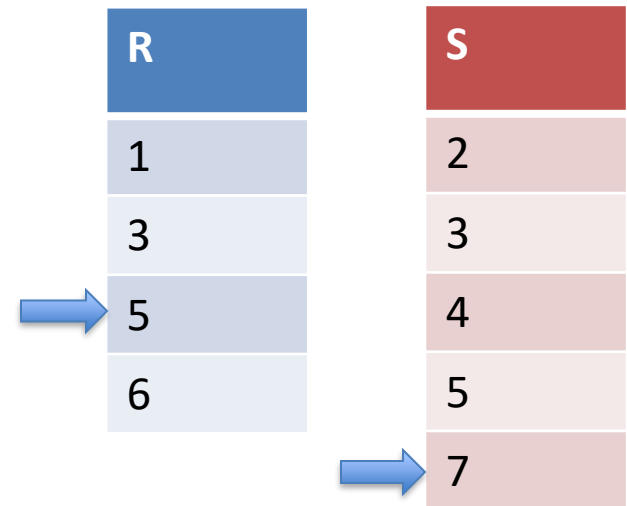
if ($R[i].\text{joinAttr} < S[j].\text{joinAttr}$):



$i = i + 1$

else:

$j = j + 1$



Output: 3, 5

Sort Merge Join

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while ($i < \{R\}$ and $j < \{S\}$):

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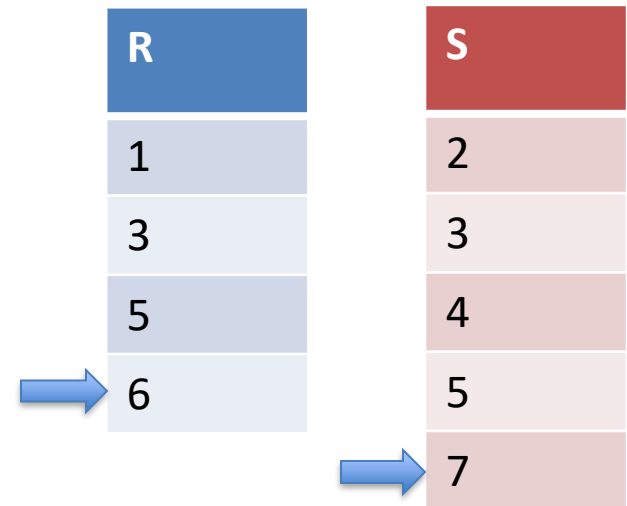
if ($R[i].\text{joinAttr} < S[j].\text{joinAttr}$):



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Output: 3, 5

Sort Merge Join

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- Merge (no shared duplicates)

while ($i < \{R\}$ and $j < \{S\}$):

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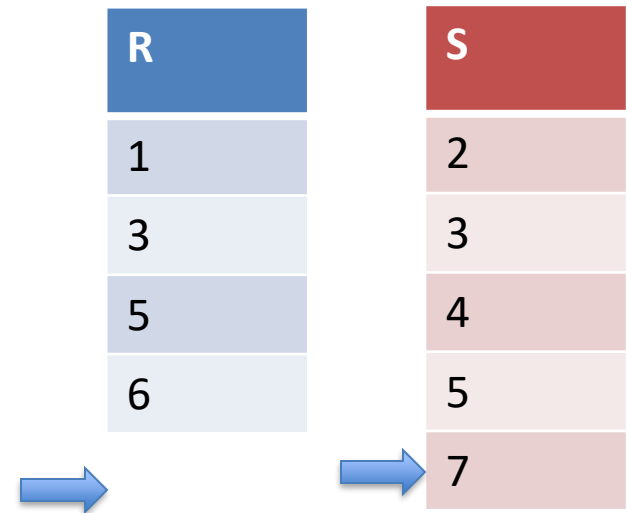
output $R[i]$ join $S[j]$

if ($R[i].\text{joinAttr} < S[j].\text{joinAttr}$):

$i = i + 1$

else:

$j = j + 1$



Output: 3, 5

Note that output is sorted!

Handling Duplicates

- What is desired output?

4 copies!

(5,5),(5,5),(5,5),(5,5)

R
1
5
5
6

S
2
3
5
5
7

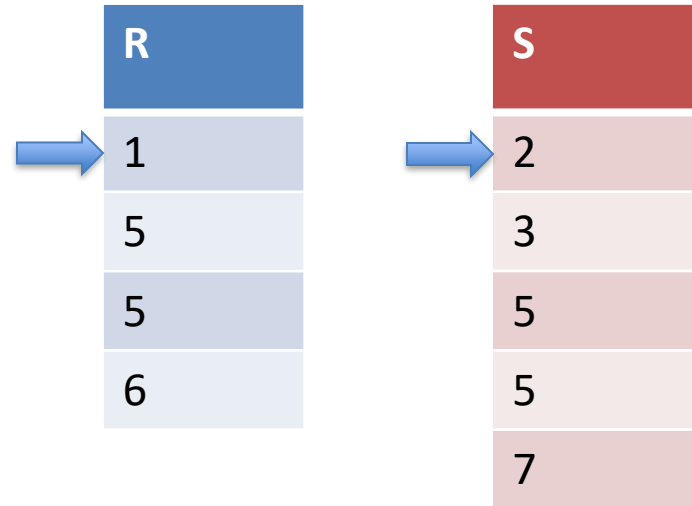
- Solution: count run lengths in S and R, emit cross product of repeated runs

Handling Duplicates

- What is desired output?

4 copies!

(5,5),(5,5),(5,5),(5,5)



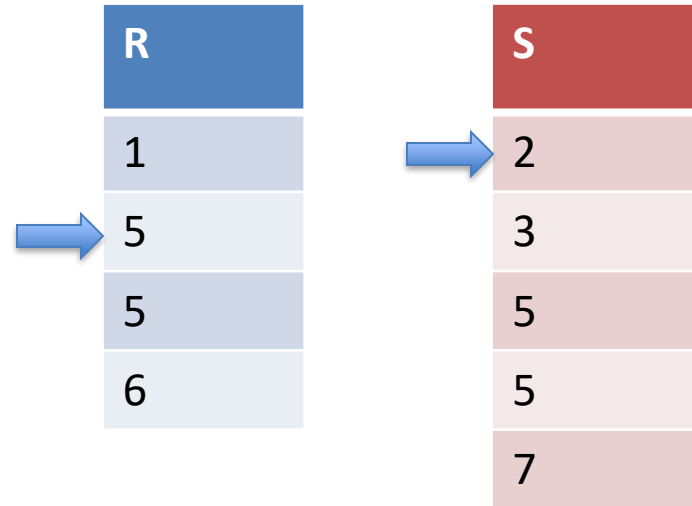
- Solution: count run lengths in S and R, emit cross product of repeated runs

Handling Duplicates

- What is desired output?

4 copies!

(5,5),(5,5),(5,5),(5,5)



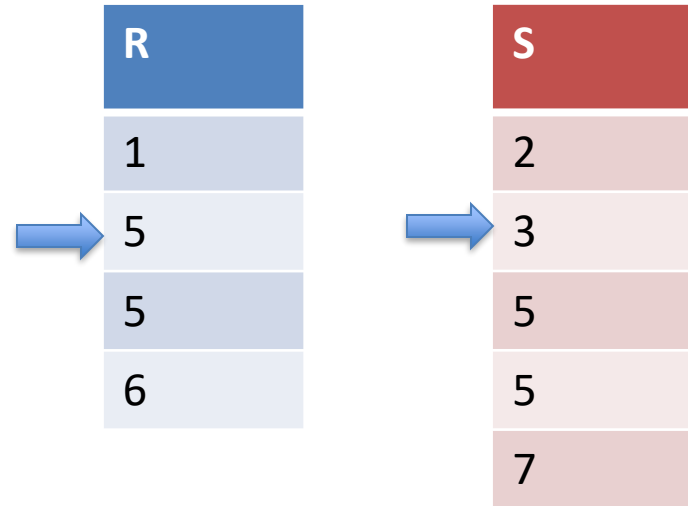
- Solution: count run lengths in S and R, emit cross product of repeated runs

Handling Duplicates

- What is desired output?

4 copies!

(5,5),(5,5),(5,5),(5,5)



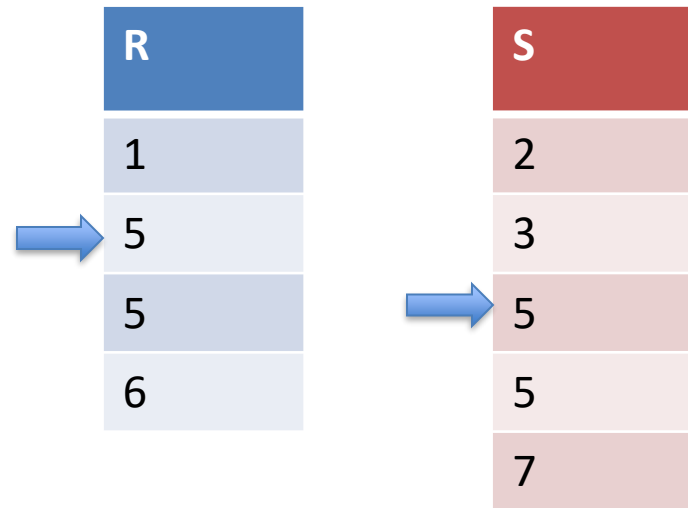
- Solution: count run lengths in S and R, emit cross product of repeated runs

Handling Duplicates

- What is desired output?

4 copies!

(5,5),(5,5),(5,5),(5,5)



Output: 5

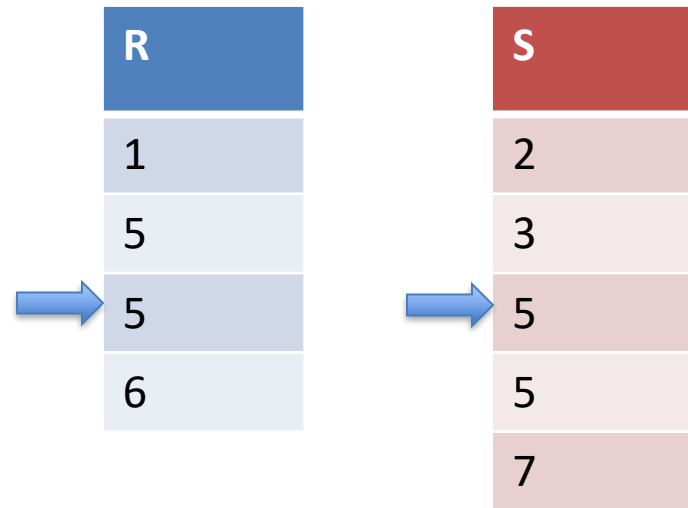
- Solution: count run lengths in S and R, emit cross product of repeated runs

Handling Duplicates

- What is desired output?

4 copies!

(5,5),(5,5),(5,5),(5,5)



Output: 5, 5

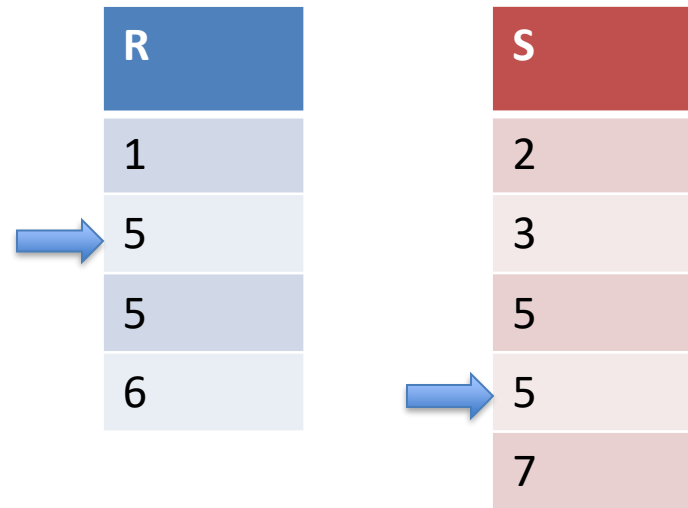
- Solution: count run lengths in S and R, emit cross product of repeated runs

Handling Duplicates

- What is desired output?

4 copies!

(5,5),(5,5),(5,5),(5,5)



Output: 5, 5

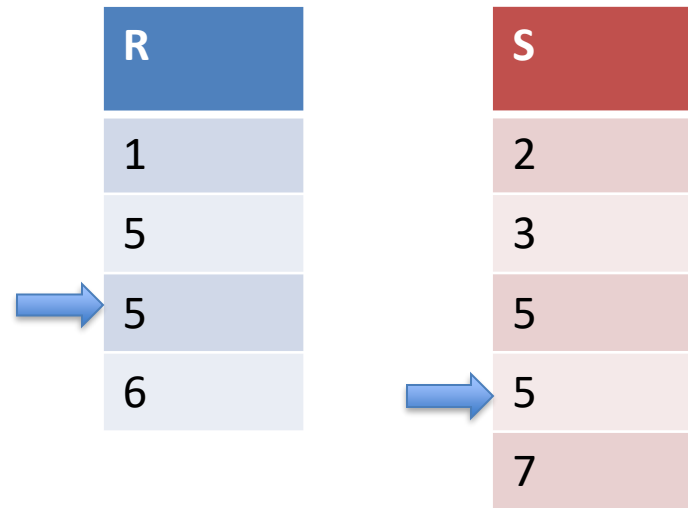
- Solution: count run lengths in S and R, emit cross product of repeated runs

Handling Duplicates

- What is desired output?

4 copies!

(5,5),(5,5),(5,5),(5,5)



Output: 5, 5, 5

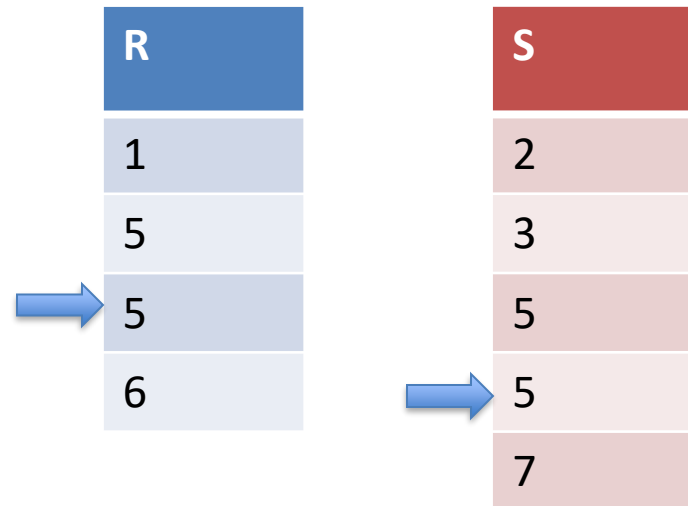
- Solution: count run lengths in S and R, emit cross product of repeated runs

Handling Duplicates

- What is desired output?

4 copies!

(5,5),(5,5),(5,5),(5,5)



Output: 5, 5, 5, 5

- Solution: count run lengths in S and R, emit cross product of repeated runs

Psuedocode for Duplicates

```
while (i < {R} and j < {S}):
    if R[i].joinAttr == S[j].joinAttr:
        rLen = getRunLen(R,i)
        sLen = getRunLen(S,j)
        emitRun(R,S,i,j,rLen,sLen)
        i = i + rLen
        j = j + sLen
    elif R[i].joinAttr < S[j].joinAttr:
        i = i + 1
    else:
        j = j + 1
```

```
def emitRun(R,S,r,s,rLen,sLen):
    for i in range(r,r+rLen):
        for j in range(s,s+sLen):
            output R[i] join S[j]
```

```
def getRunLen(v,i):
    runLen = 1
    while (i < len(v)-1):
        i = i + 1
        if v[i] == v[i-1]:
            runLen = runLen + 1
        else:
            break
    return runLen
```

Basic Join Summary

	CPU Complexity	I/O Complexity	Notes
Nested loops	$\{R\} \times \{S\}$	$ S + \{S\} R $ <i>R doesn't fit in memory</i> $ S + R $ <i>R fits in memory</i>	Choice of inner / outer matters when R fits in memory and S doesn't
Blocked nested loops	$\{R\} \times \{S\}$	$\lceil R /M \rceil \times S + R $	Better to partition R (fewer passes)
Index nested loops	$\{R\} \times D$ <i>D is tree depth, < ~5</i>	$\{R\} \times D$ <i>I/O random unless R sorted & index clustered on join attr</i>	Assuming index on S.
Hash join	$\{R\} + \{S\}$	$ R + S $	Both tables must fit in memory
Blocked hash join	$\{R\} + (\lceil R /M \rceil \times \{S\})$	$\lceil R /M \rceil \times S + R $	
Sort merge join	$\{R\}\log\{R\} + \{S\}\log\{S\} + \{S\} + \{R\}$	$ R + S $	Assumes both tables fit in memory; If already sorted, can

Study Break

- When would you prefer sort-merge over hash join?
- When would you prefer index-nested-loops join over hash join?

Join Processing in Database Systems with Large Main Memories

LEONARD D. SHAPIRO

North Dakota State University

“External” Sort Merge Join

Equi-join of two tables S & R

$|S|$ = Pages in S; $\{S\}$ = Tuples in S

$|S| \geq |R|$

M pages of memory; $M > \sqrt{|S|}$

Algorithm:

- Partition S and R into memory sized sorted runs, write out to disk
- Merge all runs simultaneously
- (Identify the smallest element from all runs; advance the pointer of the run that provided the smallest element; then test for join condition)

Total I/O cost: Read $|R|$ and $|S|$ twice, write once

$3(|R| + |S|)$ I/Os

Example

R=1,4,3,6,0,14,1,7,11

S=2,3,7

The maximum size of a run is M. The number of runs generated initially is $N = |S|/M$

R1 = 1,

To merge a set of runs, we want to keep at least one page per run in memory. That means that ideally $N \leq M$.

S1 = 2,3

(Otherwise -- if we cannot fit at least one page from each run in memory -- we need to do an intermediate sort to reduce the number of runs. Another sort run is bad, because it raises I/O costs)

PUT

R1

1



3

4

A bit of handwaving in the paper yields an outcome that we expect to avoid intermediate sorts as long as $M \geq \sqrt{|S|}$

Need enough memory to keep 1 page of each run in memory at a time

Example

R=1,4,3,6,9,14,1,7,11

S=2,3,7,12,9,8,4,15,6

R1 = 1,3,4

R2 = 6,9,14

R3 = 1,7,11

S1 = 2,3,7

S2 = 8,9,12

S3 = 4,6,15

R1	R2	R3	S1	S2	S3
1	6 	1	2 	8 	4 
3 	9	7 	3	9	6
4	14	11	7	12	15

OUTPUT

Example

R=1,4,3,6,9,14,1,7,11

S=2,3,7,12,9,8,4,15,6

R1 = 1,3,4

R2 = 6,9,14

R3 = 1,7,11

S1 = 2,3,7

S2 = 8,9,12

S3 = 4,6,15

R1	R2	R3	S1	S2	S3
1	6 	1	2	8 	4 
3 	9	7 	3 	9	6
4	14	11	7	12	15

OUTPUT

Example

R=1,4,3,6,9,14,1,7,11

S=2,3,7,12,9,8,4,15,6

R1 = 1,3,4

R2 = 6,9,14

R3 = 1,7,11

S1 = 2,3,7

S2 = 8,9,12

S3 = 4,6,15

R1	R2	R3	S1	S2	S3
1	6 	1	2	8 	4 
3 	9	7 	3 	9	6
4	14	11	7	12	15

OUTPUT
(3,3)

Example

R=1,4,3,6,9,14,1,7,11

S=2,3,7,12,9,8,4,15,6

R1 = 1,3,4

R2 = 6,9,14

R3 = 1,7,11

S1 = 2,3,7

S2 = 8,9,12

S3 = 4,6,15

R1	R2	R3	S1	S2	S3
1	6 	1	2	8 	4 
3	9	7 	3 	9	6
4 	14	11	7	12	15

OUTPUT
(3,3)
(4,4)

Example

R=1,4,3,6,9,14,1,7,11

S=2,3,7,12,9,8,4,15,6

R1 = 1,3,4

R2 = 6,9,14

R3 = 1,7,11

S1 = 2,3,7

S2 = 8,9,12

S3 = 4,6,15

R1	R2	R3	S1	S2	S3
1	6 	1	2	8 	4 
3	9	7 	3	9	6
4 	14	11	7 	12	15

OUTPUT
(3,3)
(4,4)

Example

R=1,4,3,6,9,14,1,7,11

S=2,3,7,12,9,8,4,15,6

R1 = 1,3,4

R2 = 6,9,14

R3 = 1,7,11

S1 = 2,3,7

S2 = 8,9,12

S3 = 4,6,15

R1	R2	R3	S1	S2	S3
1	6 	1	2	8 	4
3	9	7 	3	9	6 
4 	14	11	7 	12	15

OUTPUT
(3,3)
(4,4)
(6,6)

Example

R=1,4,3,6,9,14,1,7,11

S=2,3,7,12,9,8,4,15,6

R1 = 1,3,4

R2 = 6,9,14

R3 = 1,7,11

S1 = 2,3,7

S2 = 8,9,12

S3 = 4,6,15

R1	R2	R3	S1	S2	S3
1	6 	1	2	8 	4
3	9	7 	3	9	6 
4 	14	11	7 	12	15

...

OUTPUT

(3,3)

(4,4)

(6,6)

(7,7)

Output in
sorted
order!

Simple “External” Hash

Idea
Algorithm

Recall Blocked Hash Join....

Iteratively:

Build hash table on chunk of R so that hash table fits in memory

Probe (lookup in) with all of S

Repeat with next chunk of R

replace R with R', S with S'

Simple “External” Hash

Idea: Avoid repeated passes over S in blocked hash

Algorithm:

Given hash function $H(x) \rightarrow [0, \dots, P-1]$ *(e.g., $x \bmod P$)*

where P is number of partitions

for i in $[0, \dots, P-1]$:

for each r in R:

if $H(r)=i$, add r to in memory hash

otherwise, write r back to disk in R'

for each s in S:

if $H(s)=i$, lookup s in hash, output matches

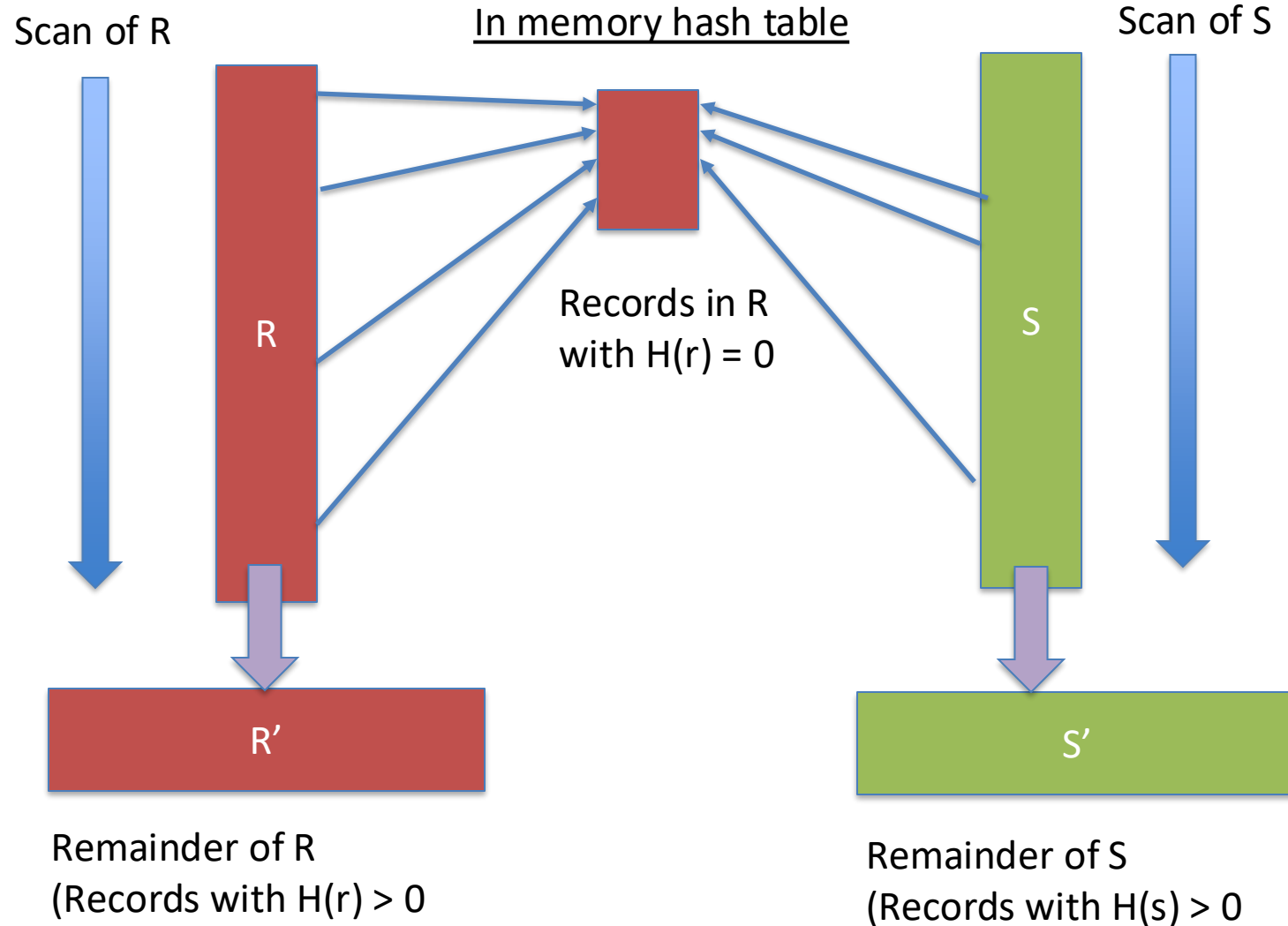
otherwise, write s back to disk in S'

replace R with R', S with S'

Pass 0

Illustration

Hash function in
 $0 \dots P$



Pass 0

Illustration

Hash function in
0...P



Pass 1

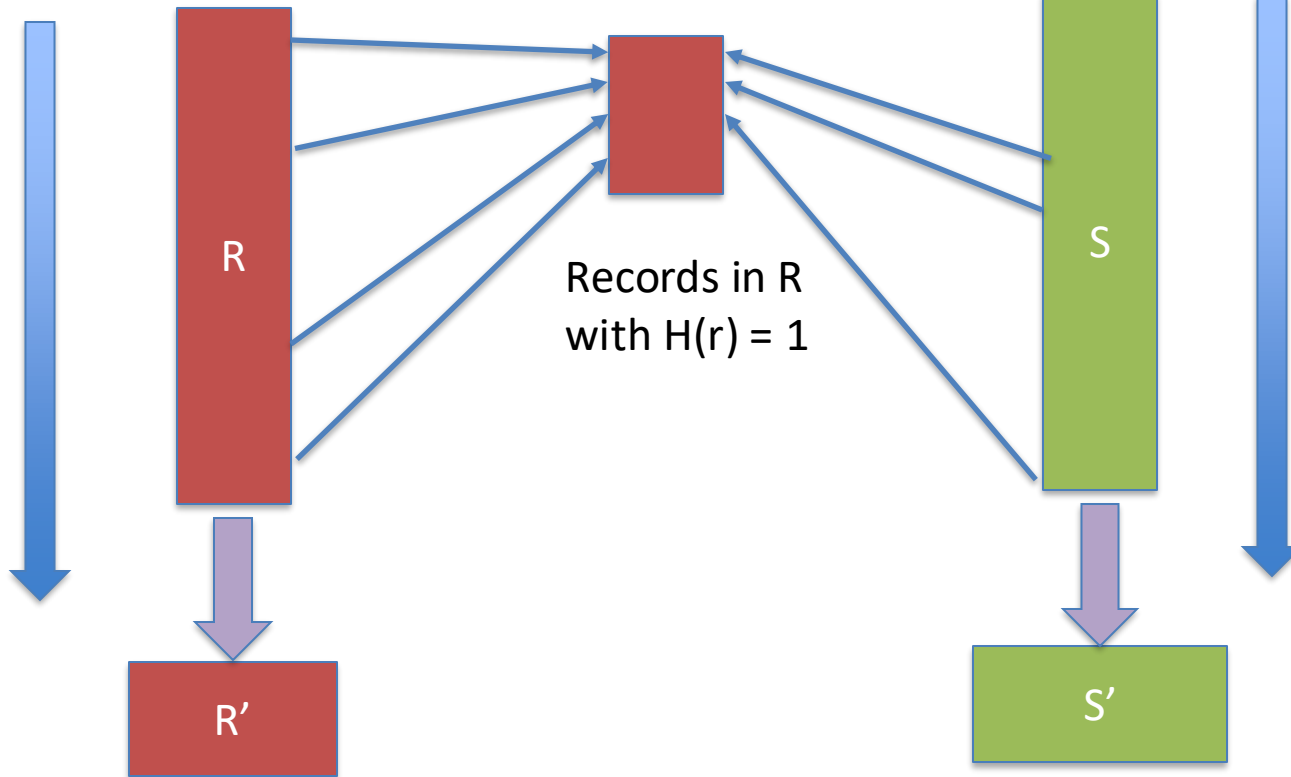
Illustration

Hash function in
 $0 \dots P$

Scan of R

In memory hash table

Scan of S



Remainder of R
(Records with $H(r) > 1$)

Remainder of S
(Records with $H(s) > 1$)

Repeat for P passes

<https://clicker.mit.edu/6.5830/>

	I/O Complexity
(A)	$n! * (R + S)$
(B)	$n * \frac{n-1}{2} * (R + S)$
(C)	$(R /n + S /n)$
(D)	$n * (R + S)$

N = number of partitions

Simple Hash I/O Analysis

Suppose $P=2$, and hash uniformly maps tuples to partitions

Read	$ R + S $	}	$2 (R + S)$
Write	$1/2 (R + S)$		
Read	$1/2 (R + S)$		

$P=3$

Read	$ R + S $	}	$3 (R + S)$
Write	$2/3 (R + S)$		
Read	$2/3 (R + S)$		
Write	$1/3 (R + S)$		
Read	$1/3 (R + S)$		

$P=4$

$$|R| + |S| + 2 * (3/4 (|R| + |S|)) + 2 * (2/4 (|R| + |S|)) + 2 * (1/4 (|R| + |S|)) \\ = 4 (|R| + |S|)$$

➔ $P = n ; n * (|R| + |S|) \text{ I/Os}$

Grace Hash

Can we avoid rewriting some records many times?

Algorithm:

Partition:

Suppose we have P partitions, and $H(x) \rightarrow [0 \dots P-1]$

Choose $P = |S| / M \rightarrow P \leq \sqrt{|S|}$ *//may need to leave a little slop for imperfect hashing*

Allocate P 1-page output buffers, and P output files for R

For each r in R :

Write r into buffer $H(r)$

If buffer full, append to file $H(r)$

Allocate P output files for S

For each s in S :

Write s into buffer $H(s)$

if buffer full, append to file $H(s)$

Join:

For i in $[0, \dots, P-1]$

Read file i of R , build hash table (*memory should hold this*)

Scan file i of S , probing into hash table and outputting matches

Total I/O cost: Read $|R|$ and $|S|$ once, write once, read back once more

$3(|R| + |S|)$ I/Os

Example

$$P = 3; H(x) = x \bmod P$$



R=5,4,3,6,9,14,1,7,11

S=2,3,7,12,9,8,4,15,6

R0	R1	R2

P output buffers

F0	F1	F2

P output files

Example

$P = 3; H(x) = x \bmod P$



$R = 5, 4, 3, 6, 9, 14, 1, 7, 11$

$S = 2, 3, 7, 12, 9, 8, 4, 15, 6$

R0	R1	R2
		5

F0	F1	F2

Example

$P = 3; H(x) = x \bmod P$



$R = 5, 4, 3, 6, 9, 14, 1, 7, 11$

$S = 2, 3, 7, 12, 9, 8, 4, 15, 6$

R0	R1	R2
	4	5

F0	F1	F2

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$S = 2, 3, 7, 12, 9, 8, 4, 15, 6$

R0	R1	R2
3	4	5
6		

F0	F1	F2

Example

$P = 3; H(x) = x \bmod P$



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$S = 2, 3, 7, 12, 9, 8, 4, 15, 6$

R0	R1	R2
3	4	5
6		

Need to flush R0 to F0!

F0	F1	F2

Example

$P = 3; H(x) = x \bmod P$



$R = 5, 4, 3, 6, 9, 14, 1, 7, 11$

$S = 2, 3, 7, 12, 9, 8, 4, 15, 6$

R0	R1	R2
	4	5

F0	F1	F2
3		
6		

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6		

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9	4	5
		14

F0	F1	F2
3		
6		

Example

$P = 3; H(x) = x \bmod P$



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R0	R1	R2
9	4	5
	1	14

F0	F1	F2
3		
6		

Example

$P = 3; H(x) = x \bmod P$



$R = 5, 4, 3, 6, 9, 14, 1, 7, 11$

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R0	R1	R2
9	4	5
	1	14

F0	F1	F2
3		
6		

Example

$P = 3; H(x) = x \bmod P$



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$S = 2, 3, 7, 12, 9, 8, 4, 15, 6$

R0	R1	R2
9		5
		14

F0	F1	F2
3	4	
6	1	

Example

$P = 3; H(x) = x \bmod P$



$R = 5, 4, 3, 6, 9, 14, 1, 7, 11$

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R0	R1	R2
9	7	5
		14

F0	F1	F2
3	4	
6	1	

Example

$P = 3; H(x) = x \bmod P$



$R = 5, 4, 3, 6, 9, 14, 1, 7, 11$

$S = 2, 3, 7, 12, 9, 8, 4, 15, 6$

R0	R1	R2
9	7	5
		14

F0	F1	F2
3	4	
6	1	

Example

$P = 3; H(x) = x \bmod P$



$R = 5, 4, 3, 6, 9, 14, 1, 7, 11$

$S = 2, 3, 7, 12, 9, 8, 4, 15, 6$

R0	R1	R2
9	7	

F0	F1	F2
3	4	5
6	1	14

Example

$P = 3; H(x) = x \bmod P$



$R = 5, 4, 3, 6, 9, 14, 1, 7, 11$

$S = 2, 3, 7, 12, 9, 8, 4, 15, 6$

R0	R1	R2
9	7	11

F0	F1	F2
3	4	5
6	1	14

Example

$P = 3; H(x) = x \bmod P$



$R = 5, 4, 3, 6, 9, 14, 1, 7, 11$

$S = 2, 3, 7, 12, 9, 8, 4, 15, 6$

R0	R1	R2

F0	F1	F2
3	4	5
6	1	14
9	7	11

Example

$$P = 3; H(x) = x \bmod P$$

R=5,4,3,6,9,14,1,7,11

S=2,3,7,12,9,8,4,15,6

R Files

F0	F1	F2
3	4	5
6	1	14
9	7	11

S Files

F0	F1	F2
3	7	2
12	4	8
9		
15		
6		

Example

$$P = 3; H(x) = x \bmod P$$

Matches:

R=5,4,3,6,9,14,1,7,11

S=2,3,7,12,9,8,4,15,6

R Files

F0	F1	F2
3	4	5
6	1	14
9	7	11

Load F0 from R into memory

S Files

F0	F1	F2
3	7	2
12	4	8
9		
15		
6		

Example

$$P = 3; H(x) = x \bmod P$$

Matches:

R=5,4,3,6,9,14,1,7,11

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9	7	11

Load F0 from R into memory

S Files



F0	F1	F2
3	7	2
12	4	8
9		
15		
6		

Scan F0 from S

Example

$$P = 3; H(x) = x \bmod P$$

Matches:
3,3

R=5,4,3,6,9,14,1,7,11

S=2,3,7,12,9,8,4,15,6

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F0	F1	F2
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F0	F1	F2
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15		
6		

Scan F0 from S

Example

$$P = 3; H(x) = x \bmod P$$

R=5,4,3,6,9,14,1,7,11

S=2,3,7,12,9,8,4,15,6

Matches:

3,3

9,9

R Files

F0	F1	F2
3	4	5
6	1	14
9	7	11

Load F0 from R into memory

S Files

F0	F1	F2
3	7	2
12	4	8
9		
15		
6		

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3,3

9,9

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3	4	5
6	1	14
9	7	11

Load F0 from R into memory

S Files

F0	F1	F2
3	7	2
12	4	8
9		
15		
6		

Scan F0 from S

Example

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S=2,3,7,12,9,8,4,15,6

Matches:

3,3

9,9

6,6

R Files

F0	F1	F2
3	4	5
6	1	14
9	7	11

Load F0 from R into memory

S Files

F0	F1	F2
3	7	2
12	4	8
9		
15		
6		

Scan F0 from S

Example

$$P = 3; H(x) = x \bmod P$$

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S=2,3,7,12,9,8,4,15,6

Matches:

3,3

9,9

6,6

R Files

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6	1	14
9	7	11

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F0	F1	F2
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Example

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Matches:

3,3

9,9

6,6

7,7

4,4

R Files

F0	F1	F2
3	4	5
6	1	14
9	7	11

S Files

F0	F1	F2
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12	4	8
9		
15		
6		

Example

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Matches:

3,3

9,9

6,6

7,7

4,4

R Files

F0	F1	F2
3	4	5
6	1	14
9	7	11

S Files

F0	F1	F2
3	7	2
12	4	8
9		
15		
6		

Hybrid

- Acts like simple for small tables, grace for large tables
- Suppose we have $M = \sqrt{|R|} + E$
 - E is additional memory beyond the minimum
- Make the first partition size E, and join as in simple
- For remaining partitions write out as in grace
- Repeat with S, joining first partition on the fly, and writing out remaining partitions as in grace
- Join remaining partitions as in grace

External Join Summary

Notation: P partitions / passes over data; assuming hash is $O(1)$

Sort-Merge	Simple Hash	Grace Hash
I/O: $3(R + S)$ CPU: $O(P \times \{S\}/P \log \{S\}/P)$	I/O: $P(R + S)$ CPU: $O(\{R\} + \{S\})$	I/O: $3(R + S)$ CPU: $O(\{R\} + \{S\})$

Grace hash is generally a safe bet, unless memory is close to size of tables, in which case simple can be preferable

Extra cost of sorting makes sort merge unattractive unless there is a way to access tables in sorted order (e.g., a clustered index), or a need to output data in sorted order (e.g., for a subsequent ORDER BY)

Many fancier versions exist, e.g., using modern sorting techniques (radix or counting sort), parallel cores, etc